American Computer Science League

2023-2024 ● Contest 3: Short Problems ● Elementary Division

1. Boolean Algebra

Of the whole numbers less than 20 (i.e. {0, 1, 2, …, 19}), how many satisfy this Boolean expression?

$$(x < 3)\ OR\ (x \geq 8)\ AND\ NOT\ (x > 15)$$

A. 8
B. 9
C. 10
D. 11
E. 12

2. Boolean Algebra

Find all of the ordered pair(s) that make the following expression TRUE:

$$\sim(A\ +\ \sim B)\ +\ \sim(\sim A\ +\ B)$$

A. (0,0), (0,1)
B. (0,1)
C. (0,1), (1,0)
D. (1,1), (0,0)
E. (1,0)

3. Boolean Algebra

Fill in the columns in the following Truth Table. Give the answer as the last column from top to bottom.

A	B	$A + B$	AB	$\sim(AB)$	$\sim(A + B)$	$\sim(AB) + \sim(A + B)$
0	0					
0	1					
1	0					
1	1					

A. 1110
B. 0001
C. 1010
D. 0110
E. 1100

4. Boolean Algebra

Simplify the following Boolean expression:

$$AB(\sim AB\ +\ \sim B\ +\ AB)$$

A. $\sim A + B$
B. AB
C. $\sim AB$
D. $A + \sim B$
E. $A\sim B$

5. Boolean Algebra

Determine which of the following is equivalent to $\sim(AB) + \sim(A + B)$:

a. $\sim A + \sim B$
b. $\sim A * \sim B$
c. $\sim(A + B)$
d. $\sim(A * B)$

A. a, c
B. b, d
C. a, d
D. a, b
E. b, c

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1. 布尔代数

在**小于** 20 **的**非负整**数**(例如{0, 1, 2, …, 19})**中**, 请问**有多少个数字满足下列布尔表达式？**

$$(x < 3)\ OR\ (x \geq 8)\ AND\ NOT\ (x > 15)$$

A. 8
B. 9
C. 10
D. 11
E. 12

2. 布尔代数

求**所有可以使得下列表达式的结果为真的有序对：**

$$\sim(A\ +\ \sim B)\ +\ \sim(\sim A\ +\ B)$$

A. (0,0), (0,1)
B. (0,1)
C. (0,1), (1,0)
D. (1,1), (0,0)
E. (1,0)

3. 布尔代数

填写**以下真值表**，将最后一列的内容按照从上到下的顺序依次写出结果。

A	B	$A\ +\ B$	AB	$\sim(AB)$	$\sim(A\ +\ B)$	$\sim(AB)\ +\ \sim(A\ +\ B)$
0	0					
0	1					
1	0					
1	1					

A. 1110
B. 0001
C. 1010
D. 0110
E. 1100

4. 布尔代数

请**简化下列布尔表达式：**

$$AB(\sim AB\ +\ \sim B\ +\ AB)$$

A. $\sim A\ +\ B$
B. AB
C. $\sim AB$
D. $A\ +\ \sim B$
E. $A\sim B$

5. 布尔代数

判断**下列哪些表达式与** $\sim(AB) + \sim(A + B)$ 等价：

a. $\sim A + \sim B$
b. $\sim A * \sim B$
c. $\sim(A + B)$
d. $\sim(A * B)$

A. a, c
B. b, d
C. a, d
D. a, b
E. b, c